OPERATOR-SYMBOLIC LANGUAGES

OSL-Core 0.5

Semantic Independence

Concise executable-language update · 01 September 2026

Komorebi P. · Core release v0.1.0

The result in one sentence

Core 0.5 keeps the Core 0.4 source grammar and receipt-v2 lifecycle, while adding separate semantic evaluators for numerical operators, contracts, sessions, metadata-facing state, and the receipt-verification intrinsic.

66/66	168/168	75/75	1,825
CORPUS PROGRAMS	TESTS PASSED	FOCUSED GATE	NATIVE RECEIPTS

STATUS

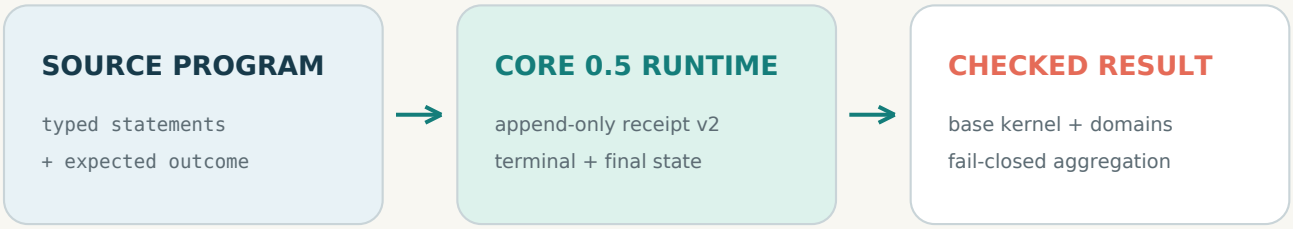
All packaged release gates pass.

Finite executable evidence—not a universal theorem or proof-assistant verification.

01 · ARCHITECTURAL DELTA

From trace to checked semantics

The new layer does not merely re-run the interpreter. Each domain report reconstructs its own receipt-facing state and is aggregated beside the protected-state kernel.



Independent coverage added in 0.5

NUM	**spaces · vectors · operators**	apply, transport, residual checks
CTR	**temporal contracts**	deadlines, violations, finalization
SES	**finite protocol graphs**	verification, advance, terminal state
META	**metadata-facing state**	secret digests, publish, show binding
INT	**receipt intrinsic**	exact-prefix verify_receipts report

WHAT REMAINS SHARED

The base kernel, metadata verifier, and intrinsic verifier still rely on primary parsing/static components. Python, NumPy, canonical serialization, and SHA-256 remain inside the trusted computing base.

Stable syntax, stricter values

Core 0.5 is an evidence-profile evolution: the source grammar remains compatible while numeric and declaration edge cases now fail explicitly.

SMALL OSL PROGRAM

```
module geo

space V dim 2

vector x in V values [1,0]

operator P : V -> V kind projector matrix [[1,0],[0,0]]

apply y = P(x)

check_projector P tolerance 1e-12

assert vector y approx [1,0] tolerance 1e-12
```

Receipt lifecycle

1. **program_open**
   exact source commitment
2. **parse + static**
   AST and report binding
3. **statement spans**
   effects and semantic deltas
4. **one terminal**
   complete or reject, last

Hardening in this release

AREA	RULE	FAILURE BEHAVIOR
Operator kinds	Allowlist: linear, projector, transport, external	Unknown kind rejected
Names	Transport results, secrets, and metadata must be unique	No silent overwrite
Numbers	Finite binary64 inputs and computed values	NaN / infinity rejected
Geometry	Shape checks; projector is a square endomorphism	No raw dimension faults
Bounds	Confidence in [0,1]; tolerances finite and ≥ 0	Explicit domain failure

Receipt schema remains osl.core.receipt.v2; core_version = 0.5 distinguishes this profile.

Five reconstructed domains

Each checker consumes source, receipts, and terminal state; recomputes domain facts; and rejects malformed, rehashed, or semantically inconsistent evidence.

DOMAIN	FRONTEND	RECONSTRUCTED SEMANTICS
Numerical	Independent fragment parser	Spaces, vectors, matrices, apply, transport, projector/intertwiner residuals, numeric rollback
Contracts	Independent source parser	All five templates, feedback deadlines, violations, terminal/finalization conditions, rollback
Sessions	Independent source parser	Protocol schema, finite graph report, start/advance/history/current/active, rollback
Metadata	Shared primary parser	Secret digests, published value/source/step/digest, assertion accounting, show semantic binding
Intrinsics	Shared primary parser	Embedded verify_receipts report over the exact preceding receipt prefix

AGGREGATION RULE

```
independent_checks_pass = AND(domain_reports)
```

A single false, missing, malformed, or inconsistent domain report makes the aggregate fail closed. The base protected-state kernel remains a parallel requirement, not a substitute.

Defined behavior at the sharp edges

The numerical fragment now has an explicit implementation profile so that runtime and independent checking can be compared without vague real-number claims.

REPRESENTATION

JSON numbers are converted to finite IEEE-754 binary64 values. Boolean values are not accepted as numbers.

LINEAR ALGEBRA

Matrix–vector multiplication, inversion, and residual norms use NumPy binary64 operations.

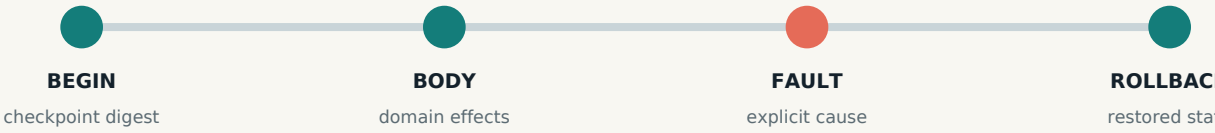
PROJECTORS

P must be a square endomorphism. The check uses $\|P^2 - P\| \le$ tolerance.

INTERTWINERS

For T, L, R, the check uses $\|TL - RT\| \le$ tolerance with finite nonnegative tolerance.

Transaction evidence



PRECISE BOUNDARY

- Numeric, contract, session, and metadata domain states are reconstructed across tested commit and rollback cases.
- The exact complete cross-domain transaction projection digest is not yet independently reconstructed end to end.

05 · FINITE EVIDENCE

What was actually run

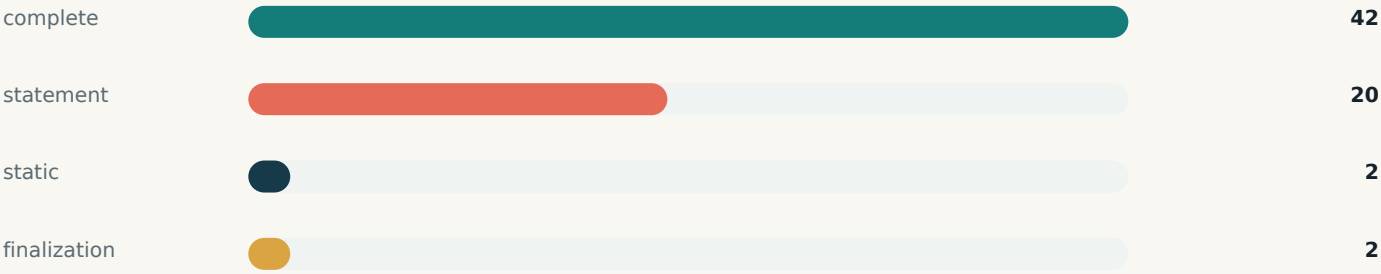
All counts below are read from the generated release reports, not projected from a plan.

66 / 66
SOURCE CORPUS

42 / 24
ACCEPT / REJECT

1,825
RECEIPT-V2 RECORDS

Terminal outcomes



GATE	RESULT	SCOPE
Full tests	168 / 168	0 failed · 0 skipped · 0 errors
Focused Core 0.5	75 / 75	five semantic domains + integration + hardening
Base independent kernel	66 / 66	lifecycle and protected state
Semantic aggregate	66 / 66	all five domain reports
Supplementary replay	66 / 66	same-runtime exact replay; explicitly secondary
Discovery Bridge	3 / 3	finite candidate paths · 71 receipts

06 · REPRODUCIBLE PACKAGE

One verifier, explicit artifacts

The archive includes the runtime, independent checkers, tests, deterministic report generators, manifest, hashes, and this technical update.

POWERSHELL

```
python -m pip install -r requirements.txt;
if ($LASTEXITCODE -ne 0) { exit $LASTEXITCODE };
.\verify_core05.ps1
```

Verification pipeline

01

Integrity

Validate manifest and SHA-256 inventory.

02

Execute

Run all 168 tests with no skips or errors.

03

Rebuild

Regenerate corpus, semantic, focused, and bridge reports.

04

Compare

Require expected outcomes and every named checker to pass.

05

Recheck

Confirm regenerated artifacts still match the package.

Offline after dependencies are installed. A failure, missing file, stale digest, or unexpected test outcome makes the release non-passing.

07 · INTERPRETATION

The strongest justified claim

Core 0.5 improves semantic separation while keeping the limits visible.

PASSING CLAIM

For the packaged finite corpus, the OSL-Core 0.5 runtime, protected-state kernel, and five domain semantic evaluators agree on the declared outcomes and satisfy the implemented receipt-v2, state-reconstruction, rollback, and language-hardening checks.

Not established

- General truth of every assertion form; metadata checks assertion accounting and selected binding, not universal truth evaluation.
- Exact complete cross-domain transaction projection digests or every rejection cause outside reconstructed domain predicates.
- Parser/static-checker correctness, Python/NumPy correctness, or proof-assistant verification of the interpreter.
- Universal safety, open-world discovery, neural semantic truth, consciousness, personhood, intent, or subjective awareness.
- Identity, authorship, consent, or nonrepudiation from SHA-256; OS sandboxing or rollback of external effects.

Next independence targets

1 **Second frontend**

Independent parse + static semantics

2 **Assertions**

Truth evaluation for all 14 forms

3 **Transactions**

Complete projection-digest reconstruction

PRIMARY RELEASE REFERENCES

README.md · CORE05_SEMANTIC_INDEPENDENCE_SPEC.md · CORE05_RECEIPT_SPEC.md · CORE05_NUMERIC_PROFILE.md · CLAIM_BOUNDARY.md · CORE05_CONFORMANCE_REPORT.json